IMSEE
Monaco Statistics

Wave climatology

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Introduction

Waves[1] are one of the main drivers of sea conditions in coastal waters. Statistical knowledge of waves (heights, periods, directions, and frequency of severe events) makes it possible to characterise a site's exposure to rough seas and storms, in support of coastal and port development projects.

This report uses measurements from buoy No. 98000 of the CANDHIS network (Centre d'archivage national de données de houle in situ, the French national archive of in situ wave data), moored off the Principality of Monaco. The network, run by Cerema (Centre d'études et d'expertise sur les risques, l'environnement, la mobilité et l'aménagement, the French centre for studies and expertise on risks, the environment, mobility, and planning), measures and disseminates data on sea states[2] along the French coastline. The data used are the buoy's half-hourly records for the last ten complete years, i.e., 2016–2025.

The study describes the site's wave climate along six lines of analysis: year-to-year changes in wave heights, the relationship between heights and periods, the identification of the most severe events, the frequency with which the 1 metre threshold is exceeded, the seasonal pattern of rough seas and their intensity, and the directional distribution of everyday waves and storms.

The annual files were first merged into a single series, missing values were processed, and the physical consistency of the measurements was checked. These steps are described in the methodological notes at the end of the report.

Temporal coverage is uneven: several years have gaps, and in three of them (2016, 2017, and 2023) these gaps fall in winter, the most energetic season of the year. The missing months were estimated so that annual totals can be compared; the values concerned are flagged throughout.

[1] Undulations of the sea surface. Strictly speaking, *swell* refers only to waves that have travelled away from the area where they were generated; in this report, as in the CANDHIS network, *waves* refer to all waves measured.
[2] Description of the sea surface over a 30-minute interval: height, period, and direction of the waves.

1. Key findings

This study characterises the wave regime measured by CANDHIS buoy No. 98000, moored off Monaco in 92 m of water, over the period 2016–2025. The half-hourly records were used to carry out six analyses: year-to-year changes in wave heights, the relationship between heights and periods, the identification of extreme events, the frequency with which the 1 m threshold is exceeded, the seasonal pattern of rough seas and their intensity, and the directional distribution of waves. Six findings emerge:

- Stable average conditions: significant wave height Hs[3] ranges on average from 0.54 to 0.67 m depending on the year, with no detectable trend over the decade.

- Waves generated in the Mediterranean: the time between two waves is short, ranging from 4.7 seconds to 6.7 seconds depending on the sea state, compared with 10 to 15 seconds for ocean swell. These values indicate that the waves are mainly generated by winds within the basin rather than arriving as swell from far away.

- Extremes from October to December: maximum Hs reached 4.51 m on 20 December 2019, together with an individual wave (HMAXD)[4] of 8.58 m, the highest in the series.

- Threshold exceedances: significant wave height exceeded 1 m on 1,165 of the 3,356 days measured, i.e. roughly one day in three. Once incomplete months are estimated, the reconstructed total is 1,274 days, an average of 127 days per year.

- Contrasting seasons: rough days are concentrated between November and March, but frequency and intensity do not peak in the same months. March is the most consistently rough month, while October to December account for all of the highest waves.

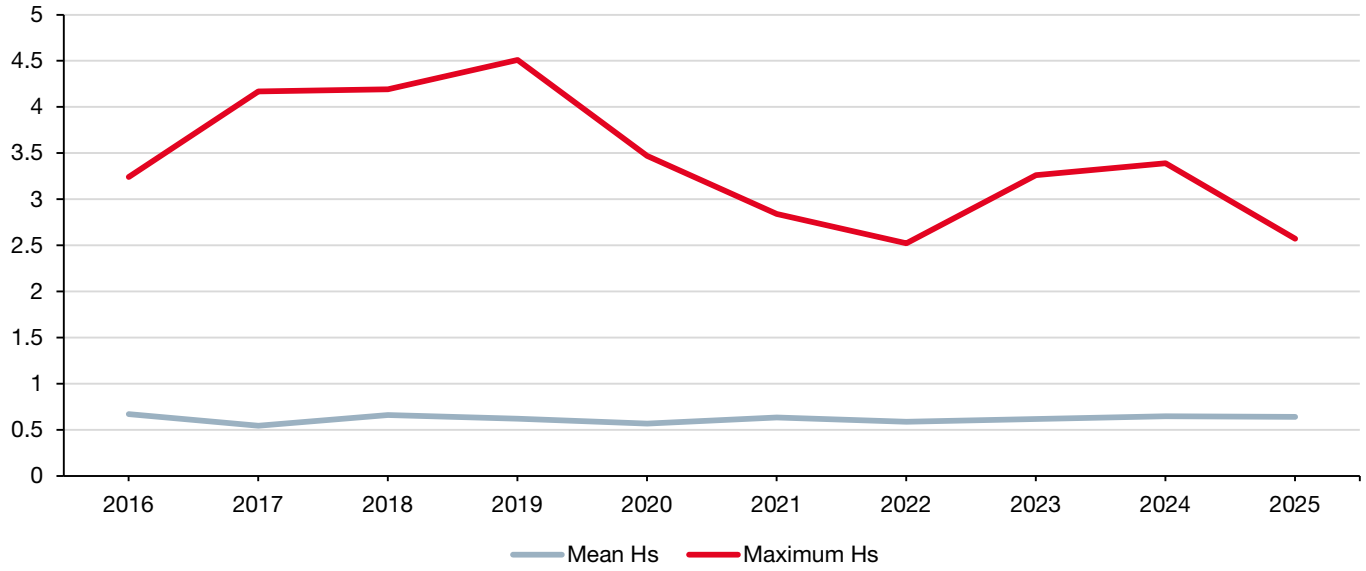
- Two dominant directions: everyday waves come mainly from the south-west (50.8%). The most severe sea states (Hs ≥ 1.5 m) come from just two sectors, south-west (54.0%) and east (33.2%), which together account for 87.2%.

[3] Height of the largest waves, close to what an observer would estimate by eye: the average of the highest third of the waves measured over 30 minutes.
[4] Highest single wave measured over 30 minutes. It can reach almost twice the significant wave height.

2. Results

2.1 Changes over the decade

Figure 1. Annual mean and maximum Hs (2016–2025)



Sources: CANDHIS archives – buoy 98000, Monaco Statistics

The chart shows two contrasting patterns. Mean significant wave height (blue line) is stable throughout the decade: it fluctuates around 0.62 m, ranging only from 0.54 m (2017) to 0.67 m (2016). Everyday sea conditions at the site therefore change very little from one year to the next.

Maximum significant wave height (red line), by contrast, is highly variable. It ranges from 2.52 m (2022) to 4.51 m (2019), almost doubling between the calmest and the roughest year. This variability is to be expected: the annual maximum depends on a single event, the strongest storm of the year.

While average conditions are stable, the intensity of extremes varies considerably from year to year. The series shows a more energetic period at the start: annual maxima peak between 2017 and 2019 (4.17 to 4.51 m), drop markedly in 2021–2022 (2.84 and 2.52 m), then recover moderately. The highest value was recorded in 2019, at 4.51 m.

Finally, the gap between the mean (0.62 m) and the maxima (2.5 to 4.5 m) highlights the intermittent nature of waves at this site: long periods of relatively calm sea, punctuated by much more intense storm episodes.

2.2 Height–period relationship

Peak period (Tp)[5] provides information on the nature of the waves. The table below gives the median[6] period observed for each significant wave height class.

Table 1. Median peak period by significant wave height class

Hs class	Median Tp (s)	Share of records
< 0.5 m	4.7	49.8%
0.5-1 m	5.5	34.1%
1-1.5 m	6.0	11.0%
≥ 1.5 m	6.7	5.1%

Note: Medians calculated from the 159,969 records with valid values for both significant wave height and peak period.

Sources: CANDHIS archives – buoy 98000, Monaco Statistics

Peak period increases with significant wave height: the median rises from 4.7 seconds for the lowest sea states (Hs < 0.5 m) to 6.7 seconds for the highest (Hs ≥ 1.5 m), in steps of around 0.7 seconds.

These values remain short compared with ocean swell, where peak periods commonly range from 10 to 15 seconds. The enclosed shape of the Mediterranean limits the distances over which the wind can act on the sea surface: waves have less room to grow in height and period before reaching the coast. For the same height, a short-period wave carries less energy, and its motion extends less far below the surface than a long swell.

[5] Time, in seconds, between two successive waves passing, for the most energetic waves.
[6] Value that divides the measurements into two equal halves.

2.3 Extreme events

To identify the most significant storms of the decade, the measurements were grouped into distinct episodes: a single storm that keeps wave heights high for several hours produces dozens of consecutive measurements, of which only the peak was retained. Two peaks are considered to belong to separate episodes when significant wave height has dropped below 1 m between them, and they are at least 48 hours apart.

Two ranking criteria were used, as they measure different things. The first, based on significant wave height Hs, reflects the overall severity of the sea state, i.e. the energy of the storm as a whole. The second, based on HMAXD, uses the highest individual wave recorded, a more localised and more extreme measure. The associated peak period is given for each episode.

Table 2. Three strongest events by Hs

Rank	Peak	Hs (m)	HMAXD (m)	Tp (s)
1	20 Dec 2019, 15:30	4.51	8.58	9.3
2	29 Oct 2018, 22:00	4.19	6.83	9.2
3	11 Dec 2017, 14:00	4.17	5.84	10.2

Note: The HMAXD column gives the maximum wave recorded at the time of the Hs peak, not the maximum reached over the whole event.

Sources: CANDHIS archives – buoy 98000, Monaco Statistics

Table 3. Three strongest events by HMAXD

Rank	Peak	Hs (m)	HMAXD (m)	Tp (s)
1	20 Dec 2019, 15:30	4.51	8.58	9.3
2	23 Nov 2019, 17:00	3.73	7.81	9.5
3	11 Dec 2017, 14:30	3.84	7.56	9.9

Sources: CANDHIS archives – buoy 98000, Monaco Statistics

Based on Hs (Table 2), the three strongest storms all occurred between October and December, in three different years: December 2019, October 2018, and December 2017. The event of 20 December 2019 stands out, with a significant wave height of 4.51 m. The next two, in October 2018 (4.19 m) and December 2017 (4.17 m), are of very similar intensity.

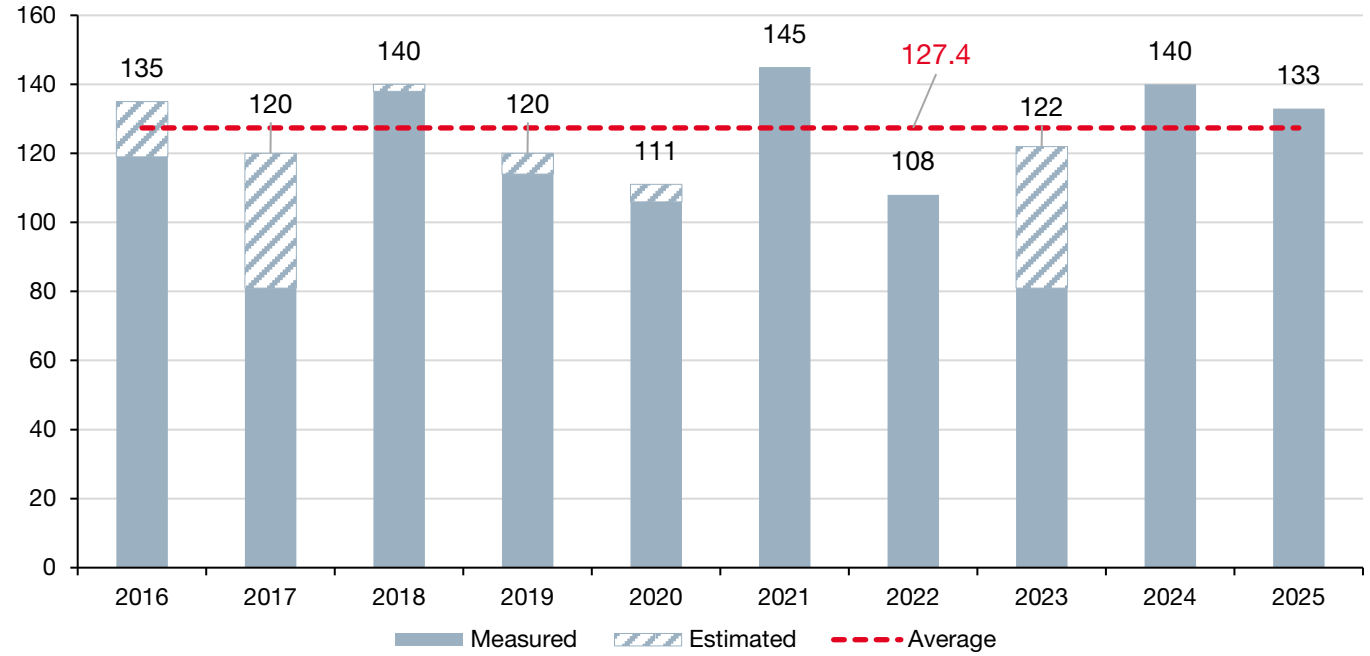
The ranking by individual wave (Table 3) largely overlaps with the first: two of the three episodes appear under both criteria, namely 20 December 2019, which ranks first in both cases with a wave of 8.58 m, the highest in the series, and 11 December 2017. The difference lies in second place, where the episode of 23 November 2019 (wave of 7.81 m) replaces that of October 2018: despite a more moderate sea state (Hs of 3.73 m), this storm produced a particularly high individual wave. A very energetic sea state and an exceptional individual wave therefore do not always coincide.

Whichever criterion is used, the major events have peak periods of 9.2 to 10.2 seconds, well above the median for energetic sea states (6.7 s for the Hs ≥ 1.5 m class). During storms, waves are therefore considerably longer than in ordinary conditions, which increases their energy for the same height.

2.4 Exceedances of the 1 m threshold

A day is counted as rough[7] when the daily maximum of significant wave height exceeds 1 m. This threshold, used as the reference throughout the study, corresponds to the level at which waves begin to affect coastal activities.

Figure 2. Annual number of days with waves above 1 m, measured and estimated



Sources: CANDHIS archives – buoy 98000, Monaco Statistics

Over the decade, significant wave height exceeded 1 m on 1,165 of the 3,356 days measured, i.e. roughly one day in three. As coverage varies from year to year, incomplete months were estimated so that annual totals can be compared; the chart distinguishes the measured and estimated share of each year. After reconstruction, the total reaches 1,274 days, an average of 127 days per year.

The roughest years were 2021 (145 days), 2018 and 2024 (140 days each). In 2016, the total reached 135 days, but this includes a December that is almost entirely estimated, as the buoy recorded only a single day that month. The same caution is therefore warranted for 2017 and 2023. At the other end of the scale, 2022 was the calmest year, with 108 days, a value based entirely on measurements.

The gaps in 2016, 2017 and 2023 fall precisely in winter, the season when rough days are most frequent. Based on raw values alone, 2017 and 2023 each showed only 81 rough days, well below all other years: they were missing January–February and November–December respectively. The case of 2016 is less visible but of the same nature.

This low count therefore reflects observation conditions, not sea conditions. Once the incomplete months are estimated, these three years reach 135, 120, and 122 days, in line with the rest of the series. They were not calmer than other years; they were simply less well observed.

Beyond this correction, no upward or downward trend in the number of rough days emerges. Active and calm years alternate with no clear direction, within a narrow range of 108 to 145 days, consistent with the stability already observed in mean wave heights.

[7] A day on which significant wave height exceeded 1 m at least once.

2.5 Seasonality - monthly distribution of rough days

Table 4. Number of days with waves above 1 m, by month and year

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total	Average
2016	17	17	20	12	18	5	2*	1*	3	7	19	14**	135	11.3
2017	15**	12**	16*	7	6	6*	6	3	10	10	11	18	120	10.0
2018	19	19	24	13	3	4	3*	2	3	20	19	11	140	11.7
2019	7	7	15	16	7	6	3	1	7*	12	21	18	120	10.0
2020	12	13	15	11	11	9	1	2	5	12*	6	14	111	9.3
2021	18	20	9	12	15	5	3	9	9	12	22	11	145	12.1
2022	9	9	18	13	2	6	2	1	15	4	16	13	108	9.0
2023	16	8	16	9	4	4	7	7*	5*	17*	15**	14**	122	10.2
2024	18	10	19	17	12	12	1	1	11	14	11	14	140	11.7
2025	23	9	20	12	8	3	6	6	9	11	13	13	133	11.1
Total	**154**	**124**	**172**	**122**	**86**	**60**	**34**	**33**	**77**	**119**	**153**	**140**	**1274**	

Note 1: Values marked with a symbol relate to months with incomplete coverage (fewer than 25 days measured) and have been estimated. A single asterisk (*) denotes a partially measured month (6 to 24 days), estimated by combining the exceedance rate observed on the available days with the average for the same month in complete years, weighted by coverage rate. A double asterisk (**) denotes a month with no usable measurements (5 days or fewer), estimated from the monthly average alone and therefore subject to greater uncertainty. These values are not observations; their purpose is to make annual totals comparable. Measured total: 1,165 days; reconstructed total: 1,274 days, i.e., 8.6% estimated.

Note 2: "Average" column. The Average column gives, for each year, the average number of days per month with waves above 1 m. Estimating the missing months allows this to be calculated over twelve months for every year.

Sources: CANDHIS archives – buoy 98000, Monaco Statistics

The table above shows, month by month and year by year, the number of days on which waves exceeded 1 m. The seasonal pattern is clear: rough days are concentrated in winter, led by March (172 days in total over the decade), January (154) and November (153), followed by December (140) and February (124). Summer is very calm, with July and August accounting for only 34 and 33 rough days respectively over ten years, about three days a month on average.

Several months stand out for exceptional activity. March 2018 (24 days with waves above 1 m out of 31), January 2025 (23 days) and November 2021 (22 days) are among the roughest in the series, with the sea exceeding 1 m on almost three days out of four. By contrast, many summer months record only one or two rough days.

2.6 Monthly intensity of extremes

Table 5. Height of the highest wave recorded, by month and year (metres)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Max
2016	4.6	5.2	5.5	4.0	4.2	3.7	3.1	2.2	3.8	3.6	5.8		5.8
2017			3.0	4.7	4.3	3.2	3.3	2.9	4.0	4.5	3.7	7.6	7.6
2018	5.4	4.2	4.5	4.7	2.9	2.6	2.5	2.4	4.5	7.2	4.1	4.6	7.2
2019	2.8	5.2	3.2	3.6	5.4	2.7	3.3	2.6	1.9	5.1	7.8	8.6	8.6
2020	5.3	4.8	5.4	4.4	4.7	3.2	2.1	2.7	3.5	4.1	3.7	6.0	6.0
2021	5.1	4.5	4.2	4.5	4.8	2.8	3.3	3.0	3.0	4.7	4.5	3.3	5.1
2022	3.1	3.4	4.9	4.7	2.5	3.3	2.0	3.0	4.0	3.4	3.7	4.3	4.9
2023	5.8	5.9	4.3	3.6	3.0	3.6	3.5	3.0	2.5	6.0			6.0
2024	5.4	4.4	5.6	4.4	4.0	3.2	1.9	2.4	3.3	4.4	5.0	4.2	5.6
2025	4.9	3.9	4.7	3.5	3.8	3.1	2.9	3.6	2.8	3.9	3.6	4.7	4.9
Max	**5.8**	**5.9**	**5.6**	**4.7**	**5.4**	**3.7**	**3.5**	**3.6**	**4.5**	**7.2**	**7.8**	**8.6**	**8.6**

Note: No values have been estimated: a maximum corresponds to a single event and cannot be reconstructed by calculation. Empty cells (December 2016; January–February 2017; November–December 2023) correspond to months with no measurements.

Sources: CANDHIS archives – buoy 98000, Monaco Statistics

This table shows, for each month, the height of the highest individual wave recorded.

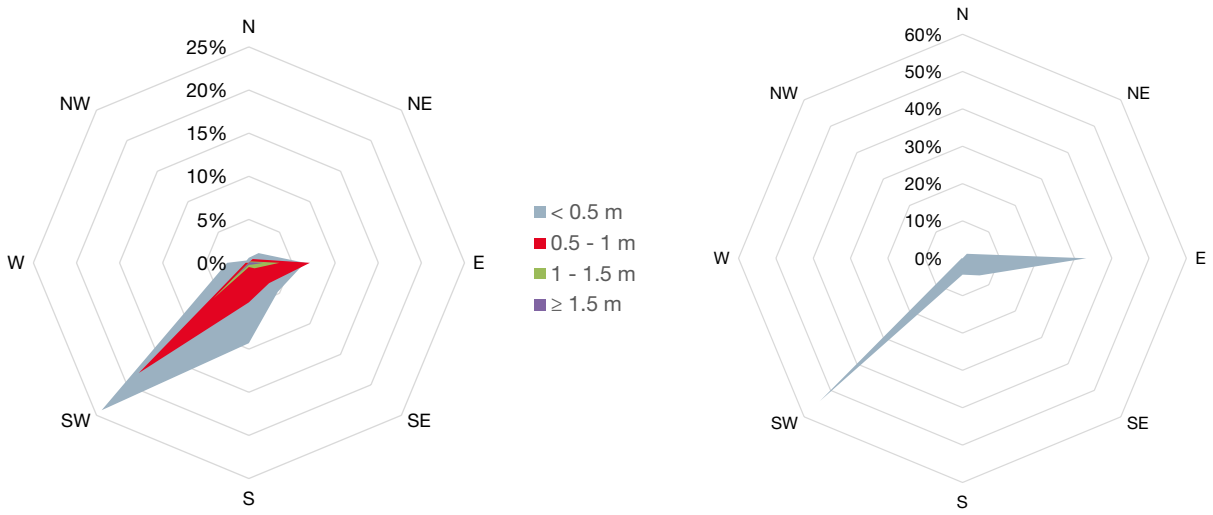
The highest wave in the series reached 8.6 m in December 2019, followed by 7.8 m in November 2019, 7.6 m in December 2017 and 7.2 m in October 2018, the four events already identified as the most severe of the decade. Maximum values in summer range from 2 to 3.6 m, with the lowest values at around 1.9 m (July 2024 and 2022). Even in the calmest season, waves of 2 to 3 m are observed every year.

Frequency and intensity do not peak at the same time of year. March has the highest number of days with waves above 1 m over the decade, yet its maximum for the decade does not exceed 5.6 m. Conversely, October to December, although less frequently rough, account for all the most violent waves (7.2 m, 7.8 m and 8.6 m). The site's wave regime thus falls into two seasons of a different nature: a winter and spring with frequent but moderate rough seas, and an autumn and early winter that are less regular but produce the extreme events.

2.7 Directional distribution

The direction[8] from which waves arrive at the peak (THETAP parameter) was divided into eight 45° sectors and cross-tabulated with significant wave height classes. Two representations are given: a wave rose[9] for all conditions, which shows where waves as a whole come from, and a storm rose restricted to the most severe sea states (Hs ≥ 1.5 m), which identifies the origin of high-energy events.

Figure 3. Wave roses: all conditions (left) and sea states ≥ 1.5 m (right)



Sources: CANDHIS archives – buoy 98000, Monaco Statistics

Under everyday conditions (left rose), waves come overwhelmingly from the south-west, which accounts for more than half of all observations (50.8%). The east forms a second well-defined direction (18.8%), followed by the south (14.6%) and the south-east (9.3%). Waves from the north, north-west and west are almost non-existent (less than 6% combined). This distribution reflects the buoy's geographical position: moored on the northern coast of the sea, it is sheltered from northerly to westerly waves by the coastline but remains exposed to the open sea to the south-west and east.

The storm regime (right rose) is more concentrated and, above all, different from the everyday regime. Severe sea states come almost exclusively from two directions, the south-west (54.0%) and the east (33.2%), which together account for 87.2% of storms. The eastern sector becomes much more prominent in high-energy conditions: its share rises from 18.8% under everyday conditions to 33.2% during storms. The south follows the opposite path, from 14.6% under average conditions to 4.4% during severe events.

[8] Direction from which the waves come, measured relative to geographic north.
[9] Circular chart showing, for each direction, the proportion of observations in which waves come from that sector.

3. Conclusion

This study characterises the wave climate at CANDHIS buoy No. 98000, off Monaco, over the period 2016–2025. Four main conclusions can be drawn.

Average conditions are stable over the decade. Mean significant wave height remains around 0.62 m from one year to the next, with no upward or downward trend, and the frequency of days with waves above 1 m, roughly one day in three, shows no drift either. The site's background level of wave activity therefore remains constant over ten years.

Year-to-year variability is driven by extreme events, not by the average. These events occur between October and December, and the intensity of the strongest storm varies considerably from year to year. The year 2019 stands out over the period, with a significant wave height of 4.51 m and an individual wave of 8.58 m, the highest in the series. Physical consistency checks confirm that the measurements are reliable at the highest values.

Frequency and intensity of rough seas diverge over the year. Days with waves above 1 m are concentrated between November and March, with March the most consistently rough month; the most violent waves, however, occur exclusively between October and December, while March has never exceeded 5.58 m in ten years. The site thus experiences two seasons of a different nature: one characterised by frequent but moderate rough seas, and the other by less frequent but more severe events.

The directional analysis reveals a bimodal distribution. Everyday waves are spread widely around a dominant south-westerly direction, but storms come almost exclusively from two sectors, south-west and east, which together account for 87.2% of the most severe sea states. This dual exposure is consistent with the site's position on the northern coast of the sea, where the coastline and terrain block waves from the north to the west.

4. Limitations

Measurement gaps are concentrated in the roughest season. The years 2016, 2017, and 2023 have gaps in winter (December 2016, January–February 2017, November–December 2023), and several other months are only partially covered. In total, 15 of the 120 months have fewer than 25 days of measurements. Without correction, these years would appear artificially calm: 2017 and 2023 each show only 81 rough days measured.

Reconstructed totals are estimates, not observations. Incomplete months were filled in using the average behaviour of the same month in complete years. This reconstruction accounts for 8.6% of the ten-year total, but for around one third of the totals for 2017 and 2023. Cross-validation[10] indicates an average error of 3.5 days for a fully reconstructed month: sufficient to compare years with one another, but not to treat any annual total as exact. The method also assumes that a missing month resembles the average; a genuinely exceptional month cannot be reconstructed.

Maximum values cannot be estimated. Unlike the number of rough days, a maximum value corresponds to a single event and cannot be reconstructed by calculation. The annual maxima for 2016, 2017 and 2023 are therefore calculated over incomplete years, missing precisely the months of October to December or January–February, when storms are concentrated. They are probably underestimated, which calls for caution when interpreting changes in the maxima and the corresponding trend test.

Extreme values describe the period observed. The ranking of events reflects the strongest storms recorded; a more intense episode occurring during a period without measurements cannot be ruled out. With ten years of data, these values are not reference values for the design of coastal structures, which would require a longer series and a dedicated statistical analysis of extremes.

[10] A validation procedure consisting of hiding known values, estimating them and then comparing the estimates with the actual values.

5. Methodological notes

5.1 Glossary

Cross-validation: A validation procedure consisting of hiding known values, estimating them and then comparing the estimates with the actual values.

Mann-Kendall test: Statistical test that checks whether a series of values tends to increase or decrease over time, without assuming that the change is regular.

Maximum wave height (HMAXD): Height of the highest single wave measured over 30 minutes. It can reach almost twice the significant wave height.

Median: Value that divides a set of measurements into two equal halves: half of the values are below it, the other half above.

Peak period (Tp): Time, in seconds, between two successive waves passing, for the most energetic waves. A short period (4 to 6 s) indicates a sea generated nearby; a long period (10 s or more) indicates swell from far away.

Rough day: Day on which significant wave height exceeded 1 m at least once.

Sea state: Description of the sea surface over a short interval, here 30 minutes: height, period, and direction of the waves. Each buoy measurement corresponds to one sea state.

Significant wave height (Hs): Height of the largest waves, close to what an observer would estimate by eye. Technically, the average of the highest third of the waves measured over 30 minutes (H13D field).

Storm: In this report, an episode during which significant wave height reaches high values; the wave roses consider sea states of at least 1.5 m.

Swell: Waves that have travelled away from the area where the wind generated them; more regular and longer than wind sea, they can travel hundreds of kilometres. In this report, as in the CANDHIS network, the broader term waves is used for all waves measured, whatever their origin.

Wave: Undulation of the sea surface, described by its height (from trough to crest) and its period (time between two crests passing).

Wave direction: Direction from which the waves come, as for wind: a south-westerly wave arrives from the south-west. It is measured relative to geographic north (THETAP parameter).

Wave rose: Circular chart showing, for each direction, the proportion of observations in which waves come from that sector.

Wind sea: Waves raised by the wind blowing over the very area where they are observed. Short and irregular, they die down quickly once the wind drops.

5.2 Temporal coverage

Measurement coverage is uneven. Most years are well documented, but three of them (2016, 2017, and 2023) have gaps in winter, the most energetic period, so their statistics should be interpreted with caution. A few isolated months in other years are also only partially covered.

Table 6. Annual coverage of Hs measurements

Year	Hs records	Coverage	Gaps identified
2016	14,959	≈ 85 %	December almost entirely missing
2017	13,236	≈ 76 %	January–February missing
2018	16,626	≈ 95 %	July partially covered
2019	16,203	≈ 92 %	September partially covered
2020	16,805	≈ 96 %	October partially covered
2021	17,502	≈ 100%	—
2022	17,510	≈ 100%	—
2023	12,429	≈ 71 %	November–December missing
2024	17,491	≈ 100 %	—
2025	17,208	≈ 98 %	—
Total	159,969		

Sources: CANDHIS archives – buoy 98000, Monaco Statistics

5.3 Data consolidation and preparation

The data come from ten annual files covering 2016 to 2025, with measurements recorded at 30-minute intervals, i.e. 17,520 records for a full year and 17,568 for a leap year (2016, 2020, and 2024). These files were merged into a single time series of 160,600 rows, after checking that headers were consistent across years, removing duplicate timestamps at the junctions between years and sorting the records chronologically.

Missing values, coded 999.999 in the source format, were then removed. This code affects the parameters unevenly: it applies extensively to the spectral fields and wave-system fields, which are not populated in this extract, but to only 631 records for significant wave height Hs. After excluding these rows, the dataset used for all analyses contains 159,969 valid Hs measurements.

5.4 Measurement site - buoy 98000 (Monaco)

Buoy 98000 is moored off the Principality of Monaco, at 43°42.815′ N / 07°25.570′ E, in 92 m of water. It is equipped with a Datawell DWR-4/ACM directional wave buoy, which measures wave heights, periods, and directions. Its location matters for interpreting the results: open to the sea to the south, the station is sheltered from northerly to westerly waves by the coastline and surrounding terrain. The Mediterranean setting explains the general characteristics observed, namely moderate heights, short periods and marked seasonality. The buoy belongs to the CANDHIS network, run by Cerema, which measures and disseminates sea-state data along the French coastline.

5.5 Identification of extreme events

The strongest events were identified by ranking the records in descending order of height, then grouping the measurements belonging to the same storm: an episode that keeps wave heights high for several hours generates dozens of consecutive measurements, of which only the peak was retained. Two peaks are assigned to separate episodes when two conditions are met: significant wave height has dropped below the 1 m threshold between them, and the peaks are at least 48 hours apart. This double condition avoids splitting a single storm into several episodes during a temporary lull. Over the study period, the episodes retained are in any case several weeks to several months apart, and the ranking does not depend on the exact criterion chosen.

5.6 Trend tests

The absence of a trend over the decade was verified statistically. Two tests were applied to the annual series: linear regression, a parametric test of the slope, and the Mann-Kendall test[11], a non-parametric test of monotonicity that is better suited to short series and insensitive to extreme values. The tests were applied to aggregated annual values rather than to the half-hourly measurements: the latter are strongly autocorrelated, as two consecutive readings are almost identical, and using them would greatly overstate the significance of the results.

[11] Statistical test that checks whether a series tends to increase or decrease over time, without assuming that the change is regular.

For mean significant wave height, neither approach detects a trend: the slope is +0.001 m/year, with p = 0.79 for the regression and p = 1.00 for the Mann-Kendall test, and a coefficient of determination close to zero (R^2 = 0.01). The finding is the same for the annual number of days with waves above 1 m (slope of +0.18 days/year; p = 0.91 and p = 1.00).

The annual maximum shows a negative slope of –0.14 m/year, which does not reach the usual significance level[12] of 5% (p = 0.064 for the regression, p = 0.21 for Mann-Kendall). This apparent decrease in extremes therefore cannot be considered established; it most likely reflects year-to-year variability in storms, since each annual maximum depends on a single event.

5.7 Limits of interpretation

Ten years are not enough to characterise a climate trend. Natural variability in Mediterranean waves occurs on multi-year to decadal timescales, and detecting a long-term signal requires several decades of observations. The results therefore describe the stability of the wave regime over the period observed, without allowing any conclusion about longer-term change.

5.8 Estimation of incomplete months

As measurement coverage is uneven, partially observed months were estimated so that annual totals can be compared. A month is considered incomplete when it has fewer than 25 days of measurements, the threshold adopted as the limit of representativeness.

Two approaches are combined depending on the degree of coverage. For partially observed months (6 to 24 days of measurements), the estimate combines the exceedance rate observed on the available days, extrapolated to the whole month, with the reference average for the same month calculated from years with complete coverage. These two terms are weighted by the coverage rate: the more a month is observed, the more its own behaviour prevails; the less it is observed, the closer the estimate comes to the month's average behaviour. For months with no usable observations (5 days or fewer), extrapolation is no longer meaningful and only the reference average is used. In all cases, the estimated value cannot be lower than the number of exceedance days actually observed.

Example: in October 2023, 12 of the 31 days were measured (38.7% coverage), 10 of which were rough. Extrapolated to the whole month, this rate would give 25.8 days. The reference average for October is 11 days. The estimate is 0.39 × 25.8 + 0.61 × 11, i.e. 17 days.

This correction applies to 15 of the 120 months and raises the ten-year total from 1,165 measured days to 1,274 reconstructed days, an estimated share of 8.6%. It affects the years very unevenly: none for four of them (2021, 2022, 2024 and 2025), while the estimated share reaches 11.9% of the annual total for 2016, 32.5% for 2017, and 33.6% for 2023.

The reliability of the imputation was assessed by cross-validation on the 105 months with complete coverage: each month was in turn treated as missing, and the estimate was compared with the actual value. For a fully missing month, the mean absolute error is 3.5 days, with almost no bias (–0.1 days). For a partially observed month, it decreases with coverage: 2.9 days at 20% coverage, 2.4 days at 40%, and 1.1 days at 80%. Compared with annual totals of between 108 and 145 days, these errors remain an order of magnitude smaller than the differences between years. The estimation therefore does not affect comparisons between years, although it does not make the reconstructed totals exact.

[12] Threshold below which a result is considered unlikely to be due to chance alone, set here at 5%.

Wave climatology

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